

# **Does M&A activity affect non-financial disclosures?**

Name, University, and Student number have been omitted for Privacy reasons

Merger and Acquisition module

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# **1 Abstract**

This research investigates the relationship between Merger and Acquisition (M&A) activity and non-financial disclosures (NFDs), with a focus on corporate ESG performance. Given the increasing emphasis on transparency following the EU's Non-financial Reporting Directive, this study utilises OLS estimations on a comprehensive dataset spanning 2010-2024. Non-financial disclosures are proxied using Refinitiv Eikon's ESG Combined Score and Environmental Score. Empirical results demonstrate a significant positive association between M&A activity and these disclosure ratings suggesting that firms engaging in deals exhibit higher levels of transparency and stakeholder engagement. This inclusion of NFRD as a regulatory shock confirms that mandatory reporting requirements have enhanced corporate disclosure quality evidenced by improved ESG scores. Furthermore, this study identifies board gender diversity and board independence are key corporate governance determinants that improve ESG performance. Results indicate firm profitability serves as a prerequisite for high-quality disclosure, whereas high capital expenditure can constrain sustainability investment. This paper illustrates how investment and corporate restructuring activity may shape corporate transparency.

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### 3 Introduction

Non-financial disclosures are an increasingly topical issue among scholars following the implementation of Directive 2014/95/EU on the disclosure of non-financial and diversity information of certain large entities or groups (the Non-financial Reporting Directive, NFRD). The NFRD pushes for greater corporate transparency and accountability regarding non-financial information within the EU to boost growth and strengthen confidence. The World Bank Group also initiated the Climate Action Plan 2021-2025 (though the plan has since been extended to June 30, 2026) which aligns with the goals of the Paris Agreement 2015, which includes measurable reductions in greenhouse gas emissions (World Bank Group, 2021). Relatedly, the International Sustainability Standards Board issued the International Financial Reporting Standards (IFRS) Sustainability Disclosure Standards: IFRS S1 General Requirements for Disclosure of Sustainability-related Financial Information, and IFRS S2 Climate-related Disclosures, highlighting the growing regulatory pressure on firms to disclose their non-financial information (ISSB, 2023b, 2023a).

In this study, the empirical data used to proxy NFDs are firm are Refinitiv Eikon's ESG Combined Score, Environmental Score, and the estimated CO<sub>2</sub> equivalents emission total. It should be noted the terms ESG (Environmental, Social, Governance) and CSR (Corporate Social Responsibility) are used interchangeably throughout as they each relate to non-financial information firms may disclose<sup>1</sup>.

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<sup>1</sup> A key distinction being ESG includes corporate governance, unlike CSR.

## 4 Literature review

There is little existing literature exploring potential association between M&A activity and non-financial disclosures (hereafter, NFDs). Given the increased implementation of mandatory non-financial disclosures (see NFRD), researchers have looked at market perceptions, Grewal, Riedl and Serafeim (2019) find the equity market perception is that mandatory disclosure will lead to net benefits for firms with strong non-financial performance and disclosure, with net costs for firms with weak non-financial performance and disclosure.

Jackson et al. (2020) assess the impact of such mandatory NFD on CSR levels using cross-country analysis, finding firms in countries with an NFD mandate have higher levels of CSR engagement, although crucially, NFD requirements have no significant impact on reducing corporate irresponsibility<sup>2</sup>. The authors articulate “self-regulation is associated with greater flexibility for businesses to develop best practices, it can also lead to complacency if firms feel no external pressure to engage with CSR” (Jackson *et al.*, 2020, p. 1), suggesting having no mandatory disclosure requirements and merely allowing firms to voluntarily disclose their non-financial information is insufficient to create positive social externalities. Furthermore, the study also suggests NFDs may be used as a method of legitimating irresponsible activities, perhaps negating the corporate accountability objective of any given mandate.

Cahan et al. (2016) find positive association between the unexpected component of NFDs and firm value<sup>3</sup>. Furthermore, the study determines the magnitude of association is greater in countries with weak nation-level institutions (such as the political system,

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<sup>2</sup> Corporate irresponsibility is defined by this study as “the set of corporate actions that negatively affects an identifiable social stakeholder’s legitimate claims” (Strike, Gao and Bansal, 2006, p. 852).

<sup>3</sup> Additionally, the findings show no association between the expected component of NFDs and firm value.

education system, and cultural system), a factor attributed to less governmental pressure to commit to an environmental agenda, therefore NFDs by firms are usually unexpected. Conversely, countries with strong nation-level institutions are more likely to have CSR mandates (see the NFRD), meaning NFDs are more often expected by the market which the evidence implies has no impact on firm valuation.

Porter & Kramer (2006, p. 80) put forth the stakeholder view of stakeholder theory, claiming “corporate success and social welfare are not a zero-sum game”, whereby social issues may be addressed by capable corporations, leading to greater competitive benefit due to the interdependency of business and society. This diverges from the traditional shareholder view, that firms should prioritise shareholder value maximisation while market forces resolve social and environmental issues (Friedman, 1970).

Empirical evidence supports the stakeholder view in this context: Wang and Wu (2024) find CSR disclosers usually face less stakeholder resistance to M&A proposals; Ma, Pan and Suardi (2024) confirm a positive relationship between ESG performance and post-merger shareholder value maximisation; Arouri, Gomes and Pukthuanthong (2019) conclude that CSR-strong firms have lower M&A associated uncertainty; and Cook et al. (2019) find firms with higher CSR performance are more profitable and valuable.

Research from China find significant positive correlation between ESG performance and shareholder value creation, the evidence reveals firms with stronger ESG performances consistently yield higher abnormal returns and exhibit superior long-run post-acquisition operating performance relative to non-disclosing counterparts (Zheng *et al.*, 2023; Ma, Pan and Suardi, 2024; Wang and Wu, 2024; Wang *et al.*, 2025). M&A's are pivotal investment activities which may be met with resistance by various stakeholders, the stakeholder view suggests firms with better CSR performance improves stakeholder support of corporate

strategy, evidenced by narrower arbitrage spreads (Arouri, Gomes and Pukthuanthong, 2019), ultimately resulting in shareholder value maximisation, implying shareholder value can be maximised by considering the welfare of other stakeholders.

Barros et al. (2022) find evidence to support the hypothesis of this study, observing M&A activity as a significant factor for increased ESG performance in the year succeeding the deal for the environmental and social components<sup>4</sup>. Additionally, the study finds larger firms are associated with higher ESG ratings in all components, noting potential explanations: firstly, the greater scrutiny placed upon larger firms from stakeholders; and secondly, the resources available (based on firm size) to improve sustainability. The latter explanation is probed by Drempetic, Klein and Zwergel (2020) who question the use of current ESG ratings and their organisational legitimacy given firm size bias and data availability, as the authors indicate non-availability is interpreted as bad news by the market. This point is furthered by Arouri, Gomes and Pukthuanthong (2019) and Dhaliwal et al. (2012) regarding the observed reduction in uncertainty and forecasting errors due to greater disclosures.

Benlemlih, Ge and Zhao (2021) explore if undervalued firms are more likely to release voluntary NFDs, the data confirms this hypothesis, mostly driven by firms initially exhibiting low CSR engagement and high stock price opacity. Evidence also finds unexpected NFD has a positive association with firm value (Cahan *et al.*, 2016), suggesting incentive for firms to release non-financial information to boost market valuation, further confirmed by Tsang, Frost and Cao (2023), who find evidence indicating firms use NFDs to manipulate investor opinions. An experimental study of commodity trading companies by Bachmann and Ingenhoff (2016) suggests firms may gain legitimacy through extensive CSR disclosures, though this legitimacy is coupled with a high degree of shareholder scepticism

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<sup>4</sup> The governance component does not appear to improve following a deal as the results were inconclusive.



due to concerns around the intent behind disclosure and psychological reactance of shareholders. Despite the limitations<sup>5</sup>, the study serves as a prospective motive for NFDs.

There is also the matter of how NFDs are communicated to the market, briefly mentioned by Bachmann and Ingenhoff (2016), who claim that it is extensive CSR disclosure which may be associated with greater firm legitimacy. Moreover, the method of disclosure has been discussed by scholars, Wang and Li (2016) conclude that the market values standalone reports of high CSR reporting quality and high perceived credibility, supported by evidence from China in which CSR disclosers have higher market valuations than matched CSR non-disclosers. Relatedly, Dhaliwal et al. (2012) find the issuance of standalone reports is associated with lower analyst forecast error<sup>6</sup>. Additionally, the association is stronger in firms where there is greater opacity in financial reporting, suggesting NFDs contribute to reduced information asymmetry, a determinant of disclosure put forth by Tsang, Frost and Cao (2023). A study of Canadian firms indicates the predominant cause for issuing standalone reports is in response to external scrutiny by stakeholders, consistent with the stakeholder view (Thorne, S. Mahoney and Manetti, 2014).

Through the M&A lens, Deng, Kang and Low (2013) determine that mergers by high CSR acquirers have faster completion times and better deal success<sup>7</sup> rates, suggesting firm CSR activity is a dominant factor in merger outcomes. Evidence from Feyisetan, Alkaraan and Le (2025) implies the probability of M&A activity is increased when ESG performance is improved, additionally, firm industry determines the degree of the increase, as financial

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<sup>5</sup> The study faces limitations regarding the measurement of shareholder scepticism, the artificial nature of the study, and a non-validated scale of corporate legitimacy. There is room for further research to accept or reject this theory.

<sup>6</sup> The findings are stronger in more stakeholder-oriented countries, where CSR performance is more closely linked with firm financial performance.

<sup>7</sup> Success in this context refers to deal completion.

firms are more likely to engage in M&A, given an increase in ESG performance relative to non-financial firms.

From another perspective, evidence finds firms with higher carbon emissions are more likely to acquire foreign targets while simultaneously being less likely to acquire domestic targets (Bose, Minnick and Shams, 2021). The explanation provided states acquiring targets from poorer foreign countries with fewer shareholder protections and weaker environmental controls is a method of outsourcing carbon risk<sup>8</sup>, especially if the domestic cost of non-compliance is high, thereby reducing the firm's financial risks. However, this outsourcing of carbon risks is perceived negatively by the market, marked by announcement returns. Another consideration is market penalties imposed on firms for their carbon emissions; however, empirical analysis finds the firms who do not disclose their emissions are further penalised for their non-disclosure (Matsumura, Prakash and Vera-Muñoz, 2014), these findings are supported by later evidence from Wang and Li (2016).

Based on the existing literature, it is expected the data will show a significant positive association between M&A activity and NFDs.

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<sup>8</sup> In the study by Bose, Minnick and Shams (2021), carbon risk is measured by emissions.

## 5 Methodology and Data

### 5.1 Methodology

To investigate the relationship between M&A activity and NFDs, the following OLS estimations are derived:

$$ESGCS = \alpha + \beta MA_{it} + \beta Controls + \theta_i + \delta_t + \varepsilon_{it}$$

$$ENESCS = \alpha + \beta MA_{it} + \beta Controls + \theta_i + \delta_t + \varepsilon_{it}$$

$$\log CO2 = \alpha + \beta MA_{it} + \beta Controls + \theta_i + \delta_t + \varepsilon_{it}$$

The dependent variables are the ESG combined score (ESGCS), ESG environmental score (ENESCS), and the logarithm of estimated CO<sub>2</sub> equivalents emissions (logCO2), described previously as the proxies for NFDs. The MA variable is binary and equals one for the year an M&A activity occurs, else a value of zero.

The controls represent the following variables added to the baseline model to control firm characteristics which can be categorised into corporate governance (board composition) and firm profitability and investment. The corporate governance variables include board gender diversity, which is the percentage of female board members, and board independence. The implications of board gender diversity have been reviewed in the literature, for example, Altunbas et al. (2022) find evidence that increasing female managers is associated with a decline in CO<sub>2</sub> emissions, while controlling for institutional differences and introducing the Paris Agreement to stress test the model. Additionally, Trinh et al. (2025) find board gender diversity reduces information asymmetry and improves internal monitoring, these are relevant considerations when discussing NFDs, hence the control variable board gender diversity is added to the baseline model. Ding et al. (2024) conclude that independent

directors are an influential factor in determining ESG performance, therefore a control for board independence is also added in relation to corporate governance.

The firm profitability and investment variables include: return on equity, the firm's return on equity measured as net income to book value of the equity presenting an accounting measure of profitability; earnings per share, earnings for the last twelve months ended the last calendar quarter, a market measure of profitability; and capital expenditure, represents the funds used to acquire fixed assets as a % of total assets. The literature, as revealed by Carroll (2015), finds that financial profitability is often a prerequisite for sustainability investments. Furthermore, Kotler and Lee (2008) find that managers strategically use ESG to improve long-term bottom-line performance through cost reductions. Earnings per share acts as a market-based metric, controlling this variable accounts for the signalling effect where firms with strong earnings use high quality non-financial disclosure to reduce information asymmetry and signal long term viability to investors (Lee, 2008). The control for capital expenditure as a percentage of total assets as high capital intensity often suggests that a firm's resources are tied up in tangible assets and operational costs which may constrain the funds available for ESG initiatives (McWilliams and Siegel, 2001).

$\theta_i$  and  $\delta_t$  represent the fixed effects for firm and year respectively, to account for the omitted variables that may be unobservable, but do not change over time.

## 5.2 Data

The purpose of this study is to determine if a relationship exists between M&A activity and NFDs made by firms. Consequently, the following hypotheses have been developed: **H0**: M&As are not associated with a change in non-financial disclosures; and **H1**: M&As are associated with a change in non-financial disclosures. This study concerns the timeframe 2010-2024, this encapsulates the implementation of the NFRD 2014 and the Paris

Agreement 2015, key events regarding NFDs. Furthermore, there is sufficient accessible data during the timeframe to conduct this study. The data contains 20,935 observed M&A events, restraints on data availability of the other variables have reduced the total sample observations in each model, however they each contain at least 11,951 observations. The independent variable in this study is M&A activity, with the data being collected from the S&P Capital IQ database.

NFDs are not easily observed on a standardised basis, therefore three proxies from the Refinitiv Eikon database are used: the ESG Combined Score (“combined score”), the Environmental Score, and the estimated CO<sub>2</sub> equivalents<sup>9</sup> emissions total (“CO2”). This data is matched with the M&A data obtained from S&P Capital IQ via their Global Industry Classification Standard, a unique code assigned to firms allowing the datasets to be combined for the purposes of this study.

Tsang, Frost and Cao (2023) note endogeneity is an issue in CSR disclosure studies due to self-selection, reverse causality, and omitted variable bias since disclosures are voluntary, therefore, the use of Eikon’s ESG scores are useful as a standardised measure across entities, though they are still based on firm disclosures which may or may not fully represent the firm’s ESG impact. The combined score pertains to ESG performance of firms based on their own reporting, however it also contains a discount for ESG controversies, scores are rated 0-100, similarly the environmental score is based on the three categories emission, innovation, and resource use, accounting for the environmental aspect of ESG scoring (LSEG Data & Analytics, 2024). The CO<sub>2</sub> variable accounts for the logarithmic estimated total CO<sub>2</sub> and CO<sub>2</sub> equivalents emission in tonnes. Additional relevant variables are added to the model relating to firm corporate governance and firm profitability. The control

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<sup>9</sup> Including the gases carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), nitrous oxide (N<sub>2</sub>O), hydrofluorocarbons (HFCs), perfluorinated compound (PFCS), sulfur hexafluoride (SF<sub>6</sub>), nitrogen trifluoride (NF<sub>3</sub>) in terms of both scope 1 (direct) and scope 2 (indirect) emissions.

group consists of firms that were not involved in M&A activity and have reported ESG ratings. Outliers were excluded from the samples due to skewness issues; the sample size remains large with over 11,000 observations in each model. The final model contains 11,999 observations for two of the dependent variables (combined score and environmental score), and 11,951 observations for the final dependent variable, CO2. Overall, the outliers were concentrated in firm profitability, where extreme values of return on equity and earnings per share exhibited very large standard deviations (95.45 and 1514.47 respectively), hence the decision to exclude extreme values while maintain a large sample size. Though it should be noted that a statistical limitation of the models presented comes from the lack of reported ESG scores due to a non-availability of firm information, a limitation also stated by Drempetic, Klein and Zwergel (2020).

## 6 Empirical results and discussion

### 6.1 Descriptive Statistics

**Table 1a**

Summary statistics

| Variables           | Obs.   | Mean    | Std. dev. | Min         | Max        |
|---------------------|--------|---------|-----------|-------------|------------|
| ESG Combined Score  | 13,253 | 44.8216 | 19.4819   | 0.7300      | 92.4100    |
| Environmental Score | 13,253 | 43.2274 | 27.0045   | 0.0000      | 98.8500    |
| CO2                 |        |         |           |             |            |
| Equivalents         | 13,755 | 12.0024 | 2.6210    | 0.6931      | 22.7555    |
| Emissions           |        |         |           |             |            |
| MA                  | 20,935 | 0.0462  | 0.2100    | 0.0000      | 1.0000     |
| <i>Controls:</i>    |        |         |           |             |            |
| Board Diversity     | 13,217 | 16.6048 | 13.7467   | 0.0000      | 100.0000   |
| Board Independence  | 13,221 | 49.0643 | 22.4432   | 0.0000      | 100.0000   |
| Return on Equity    | 20,537 | 10.2449 | 95.4527   | -7157.5500  | 8905.6100  |
| Earnings per Share  | 20,915 | 69.1002 | 1514.4710 | -43650.1000 | 72944.8000 |
| Capital Expenditure | 20,287 | 5.9144  | 10.8792   | 0.0000      | 582.6100   |

**Table 1b**

Summary statistics excluding outliers

| Variables           | Obs.   | Mean    | Std. dev. | Min      | Max       |
|---------------------|--------|---------|-----------|----------|-----------|
| ESG Combined Score  | 12,386 | 44.7139 | 19.3916   | 0.7300   | 92.4100   |
| Environmental Score | 12,386 | 43.1284 | 26.9216   | 0.0000   | 98.8500   |
| CO2                 |        |         |           |          |           |
| Equivalents         | 12,896 | 12.0060 | 2.6239    | 0.6931   | 22.7555   |
| Emissions           |        |         |           |          |           |
| MA                  | 19,754 | 0.0449  | 0.2071    | 0.0000   | 1.0000    |
| <i>Controls:</i>    |        |         |           |          |           |
| Board Diversity     | 12,374 | 16.6810 | 13.7197   | 0.0000   | 100.0000  |
| Board Independence  | 12,371 | 48.5172 | 22.2904   | 0.0000   | 100.0000  |
| Return on Equity    | 19,754 | 10.7884 | 11.6537   | -50.9600 | 60.8900   |
| Earnings per Share  | 19,754 | 9.8089  | 54.7830   | -9.6650  | 1139.8700 |
| Capital Expenditure | 19,349 | 5.9044  | 10.0835   | 0.0000   | 550.4500  |

Table 1b contains the adjusted descriptive statistics to exclude outliers, hence a smaller standard deviation in Return on Equity, and Earnings per Share.

**Table 2**

Correlation matrix

| Variables                     | (1)        | (2)        | (3)        | (4)       | (5)        | (6)        | (7)       | (8)        | (9)    |
|-------------------------------|------------|------------|------------|-----------|------------|------------|-----------|------------|--------|
| (1) ESG Combined Score        | 1.0000     |            |            |           |            |            |           |            |        |
| (2) Environmental Score       | 0.8319***  | 1.0000     |            |           |            |            |           |            |        |
| (3) CO2 Equivalents Emissions | 0.1855***  | 0.2833***  | 1.0000     |           |            |            |           |            |        |
| (4) MA                        | 0.0788***  | 0.0625***  | 0.0131     | 1.0000    |            |            |           |            |        |
| (5) Board Diversity           | 0.2754***  | 0.1890***  | -0.1195*** | 0.0474*** | 1.0000     |            |           |            |        |
| (6) Board Independence        | 0.2382***  | 0.0821***  | -0.0149*   | 0.1130*** | 0.3452***  | 1.0000     |           |            |        |
| (7) Return on Equity          | 0.0417***  | 0.0221**   | 0.0245***  | 0.0360*** | 0.0373***  | 0.0559***  | 1.0000    |            |        |
| (8) Earnings per Share        | 0.0741***  | 0.1051***  | 0.0449***  | 0.0670*** | -0.1150*** | -0.1116*** | 0.0121*   | 1.0000     |        |
| (9) Capital Expenditure       | -0.0835*** | -0.0695*** | 0.1714***  | -0.0131*  | -0.0357*** | 0.0438***  | 0.0944*** | -0.0200*** | 1.0000 |

The symbols \*, \*\*, and \*\*\* represent significant levels of 10%, 5%, and 1% respectively.



Table 2 indicates a significant relationship at least at the 10% level amongst all variables. The strongest correlation is observed between the two dependent variables combined score and environmental score, an expected relationship as the combined score contains the same data as the environmental score. Importantly, the correlations between the other regressors are not particularly large, hence we do not detect multicollinearity.

## 6.2 Models

**Table 3**

Baseline model

|                | (A)<br>ESG Combined<br>Score | (B)<br>Environmental<br>Score | (C)<br>CO2 Equivalents<br>Emissions |
|----------------|------------------------------|-------------------------------|-------------------------------------|
| Variables      |                              |                               |                                     |
| MA             | 5.9377***<br>(0.6751)        | 6.5440***<br>(0.9383)         | 0.1367<br>(0.0918)                  |
| Constant       | 44.2906***<br>(0.1802)       | 42.6619***<br>(0.2505)        | 11.9967***<br>(0.0239)              |
| Obs.           | 12386                        | 12386                         | 12896                               |
| R <sup>2</sup> | 0.00621                      | 0.00391                       | 0.000172                            |

Standard errors in parentheses

The symbols \*, \*\*, and \*\*\* represent significant levels of 10%, 5%, and 1% respectively.

**Table 4**

Corporate governance variables

|                 | (A)<br>ESG Combined<br>Score | (B)<br>Environmenta<br>l Score | (C)<br>CO<br>2<br>Equ<br>ival<br>ents<br>Emi<br>ssio<br>ns |
|-----------------|------------------------------|--------------------------------|------------------------------------------------------------|
| Variables       |                              |                                |                                                            |
| MA              | 3.8237***<br>(0.6466)        | 5.4693***<br>(0.9285)          | 0.1351<br>(0.0923)                                         |
| Board Diversity | 0.3093***                    | 0.3567***                      | -0.0250***                                                 |

|                    |            |            |           |
|--------------------|------------|------------|-----------|
|                    | (0.0128)   | (0.0184)   | (0.0018)  |
| Board Independence | 0.1365***  | 0.0162     | 0.0034*** |
|                    | (0.0079)   | (0.0114)   | (0.0011)  |
| Constant           | 32.6701*** | 36.0056*** | 12.2816** |
|                    | (0.4043)   | (0.5807)   | (0.00576) |
| Obs.               | 12371      | 12371      | 12323     |
| R <sup>2</sup>     | 0.102      | 0.0387     | 0.0152    |

Standard errors in parentheses

The symbols \*, \*\*, and \*\*\* represent significant levels of 10%, 5%, and 1% respectively.

**Table 5**

Firm profitability and investment variables

| Variables           | (A)<br>ESG Combined<br>Score | (B)<br>Environmenta<br>l Score | (C)<br>CO2 Equivalents<br>Emissions |
|---------------------|------------------------------|--------------------------------|-------------------------------------|
| MA                  | 5.6597***<br>(0.6835)        | 5.8844***<br>(0.9490)          | 0.1419<br>(0.0922)                  |
| Return on Equity    | 0.0748***<br>(0.0144)        | 0.0571***<br>(0.0200)          | 0.0012<br>(0.0019)                  |
| Earnings per Share  | 0.0228***<br>(0.0028)        | 0.0464***<br>(0.0039)          | 0.0026***<br>(0.0004)               |
| Capital Expenditure | -0.2819***<br>(0.0289)       | -0.3152***<br>(0.0402)         | 0.0658***<br>(0.0034)               |
| Constant            | 44.3684***<br>(0.2685)       | 42.6381***<br>(0.3728)         | 11.6752***<br>(0.0349)              |
| Obs.                | 12014                        | 12014                          | 12524                               |
| R <sup>2</sup>      | 0.0212                       | 0.0211                         | 0.0333                              |

Standard errors in parentheses

The symbols \*, \*\*, and \*\*\* represent significant levels of 10%, 5%, and 1% respectively.

The baseline model is presented in Table 3; the three dependent variables are regressed against the M&A binary variable. The coefficient of the M&A variable is positive

and significant for both the combined score (5.9377) and environmental score (6.5440), indicating firms that engaged in M&A exhibited higher ESG ratings than the control group. However, while the coefficient for CO<sub>2</sub> is positive (0.1367), it is not significant in this model, suggesting M&A activity does not have a significant impact on emissions, at least in the short term. This may be due to limitations regarding the speed at which a firm can realistically implement carbon reducing initiatives internally and through its supply chain, therefore the long-term trend in CO<sub>2</sub> emissions following an M&A event may be a topic for further research. This model does not hold firm characteristics constant, acquiring firms tend to be larger with better profitability ratios, monitoring mechanisms, and superior management techniques which may explain their better ESG score instead of the deal (M&A activity) itself (Cook *et al.*, 2019).

The impact of corporate governance factors is added to the baseline model in Table 4 where a similar pattern emerges, M&A activity is associated with a positive significant increase in combined score (3.8237) and environmental score (5.4693), whereas the coefficient for CO<sub>2</sub> (0.1351) remains positive but insignificant, *ceteris paribus*. Reviewing the coefficients of the control variables indicates board diversity is a significant factor in enhancing ESG ratings, while also significant in reducing the CO<sub>2</sub> emissions of a firm, consistent with the reviewed literature (Altunbas *et al.*, 2022). The results show board independence is significant in impacting the combined score; however, it is not significant in impacting the environmental score of a firm. This indicates it is the other components of ESG (likely the governance component) that board independence enhances, rather than the firm's environmental impact.

Secondly, the firm profitability and investment control variables are added to the baseline model in Table 5, where similar trends emerge regarding the positive and significant

coefficients of M&A activity regarding the combined score and environmental score (5.6597 and 5.8844 respectively), and an insignificant impact on CO<sub>2</sub> emissions (coefficient is 0.1146). The results show a positive and significant relationship between return on equity and earnings per share, suggesting greater profitability (from an accounting lens and a market lens) is associated with a higher ESG rating, while earnings per share is simultaneously associated with greater CO<sub>2</sub> emissions. The results show greater capital expenditure, holding all else constant, is associated with a lower ESG rating and greater carbon emissions, consistent with McWilliams and Siegel (2001).

**Table 6**  
Full model

| Variables              | (A)<br>ESG Combined<br>Score | (B)<br>Environmental Score | (C)<br>CO2 Equivalents<br>Emissions |
|------------------------|------------------------------|----------------------------|-------------------------------------|
| MA                     | 3.2817***<br>(0.6534)        | 4.5473***<br>(0.9390)      | 0.1146<br>(0.0922)                  |
| Board Diversity        | 0.3039***<br>(0.0129)        | 0.3518***<br>(0.0186)      | -0.0208***<br>(0.0018)              |
| Board<br>Independence  | 0.1524***<br>(0.0081)        | 0.0362***<br>(0.0116)      | 0.0036***<br>(0.0011)               |
| Return on Equity       | 0.0454***<br>(0.0137)        | 0.0368<br>(0.0197)         | 0.0003<br>(0.0019)                  |
| Earnings per<br>Share  | 0.0375***<br>(0.0027)        | 0.0570***<br>(0.0039)      | 0.0021***<br>(0.0004)               |
| Capital<br>Expenditure | -0.2720***<br>(0.0275)       | -0.2864***<br>(0.0395)     | 0.0898***<br>(0.0039)               |
| Constant               | 32.2238***<br>(0.4463)       | 35.0924***<br>(0.6414)     | 11.7858***<br>(0.0628)              |

|                |       |        |        |
|----------------|-------|--------|--------|
| Firms FE       | YES   | YES    | YES    |
| Year FE        | YES   | YES    | YES    |
| Obs.           | 11999 | 11999  | 11951  |
| R <sup>2</sup> | 0.120 | 0.0570 | 0.0588 |

Standard errors in parentheses

The symbols \*, \*\*, and \*\*\* represent significant levels of 10%, 5%, and 1% respectively.

The full model presented in Table 6 amalgamates all control variables, building on the previous models. It is clear from Table 6 that it is the corporate governance variables that have the greatest influence on the combined score and environmental score, whereby an increase in gender diversity and board independence holding all else equal, is estimated to improve both scores. Additionally, it is the corporate governance variables that affect CO<sub>2</sub> the most, though significant, the coefficients are small, indicating little impact overall. Importantly, the signs of the coefficients of these variables diverge, with increased gender diversity implicating a reduction in CO<sub>2</sub> emissions, whereas increased independence of the board is associated with greater CO<sub>2</sub> emissions, though again, the magnitude of the coefficients is little (-0.0191 and 0.0034, respectively). This model supports the rejection of H<sub>0</sub>, the evidence suggests there is a relationship between M&A activity and NFDs when using the ESG combined score and environmental score as proxies.

### 6.3 Robust Model

**Table 7**

Robust model: Addition of Non-Financial Reporting Directive

| Variables             | (A)<br>ESG Combined<br>Score | (B)<br>Environmental Score | (C)<br>CO <sub>2</sub> Equivalents<br>Emissions |
|-----------------------|------------------------------|----------------------------|-------------------------------------------------|
| MA                    | 0.9791<br>(1.2535)           | 0.5869<br>(1.8018)         | -0.1071<br>(0.1780)                             |
| NFRD                  | 3.6834***<br>(0.4820)        | 4.0224***<br>(0.6928)      | -0.2043***<br>(0.0684)                          |
| MA*NFRD               | 3.8289***<br>(1.4581)        | 6.2313***<br>(2.0959)      | 0.3040<br>(0.2069)                              |
| Board Diversity       | 0.2801***<br>(0.0131)        | 0.3267***<br>(0.0188)      | -0.0191***<br>(0.0019)                          |
| Board<br>Independence | 0.1595***                    | 0.0443***                  | 0.0034***                                       |

|                     |            |            |            |
|---------------------|------------|------------|------------|
|                     | (0.0079)   | (0.0114)   | (0.0011)   |
| Return on Equity    | 0.0020     | 0.0012     | 0.0003*    |
|                     | (0.0013)   | (0.0019)   | (0.0002)   |
| Earnings per Share  | 0.0358***  | 0.0543***  | 0.0022***  |
|                     | (0.0026)   | (0.0037)   | (0.0004)   |
| Capital Expenditure | -0.2419*** | -0.2544*** | 0.0849***  |
|                     | (0.0266)   | (0.0382)   | (0.0038)   |
| Constant            | 29.4414*** | 31.8675*** | 11.9526*** |
|                     | (0.5895)   | (0.8473)   | (0.0835)   |
| Firms FE            | YES        | YES        | YES        |
| Year FE             | YES        | YES        | YES        |
| Obs.                | 12285      | 12285      | 12236      |
| R <sup>2</sup>      | 0.127      | 0.0627     | 0.0571     |

Standard errors in parentheses

The symbols \*, \*\*, and \*\*\* represent significant levels of 10%, 5%, and 1% respectively.

During the sampled timeframe, several key developments regarding NFDs emerged, most notably the EU's Non-Financial Reporting Directive (NFRD) in 2014 and the Paris Agreement (2015). Table 7 adds a new binary variable ("NFRD") to the full model which takes a value of one for the years following the implementation of NFRD requirements, this aims to test the validity of the full model and determine if the previously described positive relationship between M&A activity and NFDs is indeed robust, or if the inclusion of external factors such as the NFRD invalidates the results.

The inclusion of the NFRD binary variable and the interaction variable ("M&A\*NFRD") has produced insignificant coefficients for M&A activity regarding ESG ratings and carbon emissions. Table 7 shows the implementation of the NFRD has significant explanatory power and is associated with increases in ESG rating and decreases in CO<sub>2</sub> emissions. Holding all else constant, in the years following the NFRD, the ESG combined score of firms is 3.6834 points greater than the years before the NFRD. The interaction variable (MA\*NFRD) shows firms that engaged in M&A activity in a year succeeding the

NFRD are estimated to score 3.8289 points higher in the combined score, and 6.2313 points higher in the environmental score than the control group, significant at the 1% level, *ceteris paribus*. This suggests that the findings in previous models do not hold in all time periods, rather the observed relationship is evident only in the years following the NFRD. Given the sample timeframe (2010-2024) mostly contains the years after the NFRD was introduced, it is unsurprising that the models presented in Table 3, Table 4, Table 5, and Table 6 presented significant results for the M&A variable.

This paper seeks to explore the potential relationship between M&A activity and the NFDs made by firms, a relationship currently overlooked in the literature, though relevant given the increase in disclosure requirements in recent years (NFRD and the Paris Agreement). After stress-testing the full model with three dependent variables acting as proxies for NFDs, the results indicate a positive and significant relationship between M&A activity and NFDs, rejecting H0 in favour of accepting H1.

## **7 Conclusions and implications**

This paper explores the impact of M&A activity on non-financial disclosures observing 20,935 M&A events between 2010 and 2024. The findings reveal a consistent positive effect on ESG combined and environmental score, however no real significant reduction in CO2 emissions. As shown in the robust model, this impact is attributed to the regulatory pressure of the EUs non-financial reporting directive (NFRD). Post incorporation of the NFRD into the regression, the relationship between M&A activity and disclosure Directive suggests that policy intervention can enhance the positive effects of corporate strategy on sustainability performance. These findings have important implications for policymakers and regulators; the results support continued emphasis on disclosure frameworks to drive ESG improvements. For managers and investors, the findings suggest that M&A activity can act as a catalyst for stronger ESG scores. Further implications of these results could be realised in future M&A deals as corporate restructuring under the regulatory pressure of the NFRD may enhance non-financial disclosure practices.



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